

Mathematical Content

Math Department

April 13, 2026

1 Inline Mathematics

The quadratic formula is $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. Einstein's famous equation $E = mc^2$ relates energy and mass. The area of a circle with radius r is $A = \pi r^2$.

2 Display Equations

The Gaussian integral:

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi} \quad (1)$$

Maxwell's equations in differential form:

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0} \quad (2)$$

$$\nabla \cdot \mathbf{B} = 0 \quad (3)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad (4)$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \quad (5)$$

3 Matrix Notation

A 3x3 identity matrix:

$$I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (6)$$

4 Summation and Limits

The Taylor series expansion:

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots \quad (7)$$

The definition of a limit:

$$\lim_{x \rightarrow a} f(x) = L \tag{8}$$